

Lecture 2: Asymptotics

In this lecture, we will discuss **asymptotics**: what happens in the limit as the sample size grows large. These results are very useful because often we can say more about the behavior of a test or estimator in a large sample than in a small sample. For example, we often do not know the exact distribution of an estimator in a finite sample, but we can say that in a large sample it is normally distributed with a particular mean and variance. We will be particularly interested in the **Law of Large Numbers (LLN)** and the **Central Limit Theorem (CLT)**, which show that sample means are consistent and asymptotically normal. We will start by showing these in simulation, then we will discuss each theorem somewhat more formally. We will conclude by discussing the **Continuous Mapping Theorem** and **Slutsky's Theorem**, which will allow us later to use the LLN and CLT to study asymptotics for other estimators.

Note: Some of the material in this lecture is quite technical. In particular, the formal definitions of convergence in probability and distribution are beyond the scope of this course: I will show them for those of you who are interested, but you are only responsible for understanding the "intuitive definition" for each of these concepts. Moreover, although I will show the continuous mapping theorem and Slutsky's Theorem, since they will be useful for proving that estimators are consistent and asymptotically normal, you will not be responsible for them on any problem set, quiz, or exam.

The Law of Large Numbers

The sample mean is an essential quantity in statistics. Sometimes we are interested directly in the sample mean. Other times, we are interested in estimators that are themselves functions of sample means. Fortunately, statistics provides us with some powerful results on the sample mean.

To think about the sample mean, let us think about a sequence of random variables, $\{X_i\}_{i=1}^n$. Each X_i represents an observation in a data set with n observations. We will assume that X_i is distributed i.i.d., meaning that each draw of X_i is independent from the other observations, and is drawn from the same distribution. We will use $\bar{X}_n$ to refer to the sample mean, $\frac{1}{n} \sum_{i=1}^n X_i$.

What happens as the sample size, n , grows large? We will start by looking at some simulation results. For each simulation, we will draw from a uniform distribution, $X_i \sim [0, 1]$. For a given simulation, we draw the full sample data, $\{X_i\}_{i=1}^n$, and then take the sample mean, $\bar{X}_n$. We repeat the simulation 10,000 times, to ensure that our simulation has enough draws to accurately approximate the true distribution.

Figure 1 plots the simulated distribution of $\bar{X}_n$, for $n = 10$, $n = 100$, $n = 1,000$, and $n = 10,000$. We notice immediately that as the sample size grows large, the distribution concentrates more and more around a single point. In particular, this point is $\bar{X}_n = 0.5$, which is the true population mean. It turns out that this is not just a feature of this simulation, but a very general property of the sample mean.

To formalize this, we must first define the notion of **convergence in probability**. We will provide a formal definition and an intuitive definition: you are only responsible for understanding the intuitive definition. Let $\{X_n\}_{n=1}^\infty$ denote a sequence of random variables, and let c denote a constant.¹

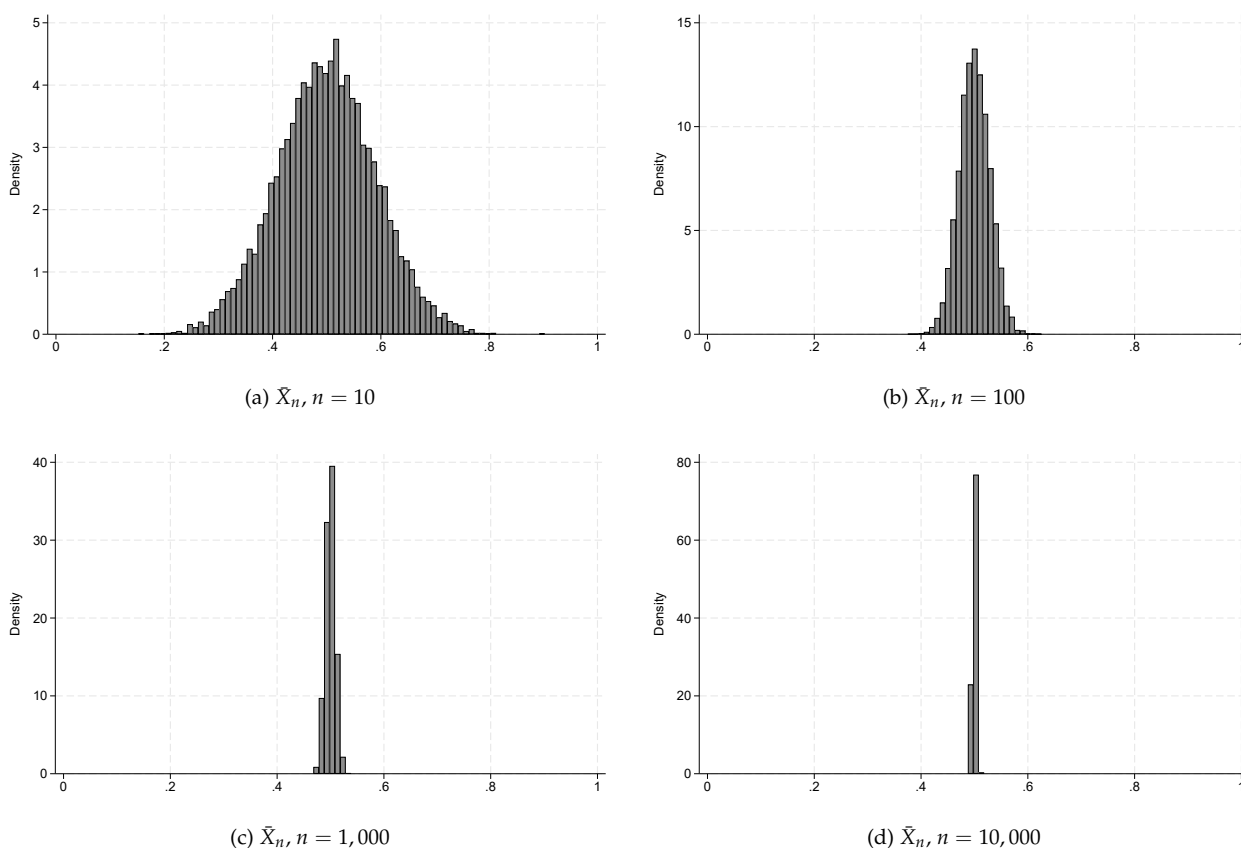
Formal Definition. We say that the sequence X_n converges in probability to the constant c if, for all $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|X_n - c| > \epsilon) = 0$.

Intuitive Definition. We say that the sequence X_n converges in probability to the constant c if, as n grows large, the distribution of X_n concentrates arbitrarily tightly around c .

We can write this as $X_n \xrightarrow{P} c$, or as $\text{plim}_{n \rightarrow \infty} X_n = c$.

Closely related to convergence in probability is **consistency**, which is a desirable property for an estimator to have. Consider a sequence of estimators $\{\hat{\theta}_n\}_{n=1}^\infty$, where $\hat{\theta}_n$ is an estimator for θ that was estimated on the first n observations in the data. Estimators are functions of the data, and they are thus random variables that can converge as n grows large. Ideally, our estimator would converge to the true parameter.

¹It is also possible for X_n to converge in probability to another random variable, X . However, we will not use that in this course, so I am omitting it here.

Figure 1: Distribution of Sample Mean, $\bar{X}_n$, for Different Sample Sizes, n

Definition. We say that $\hat{\theta}_n$ is consistent if $\hat{\theta}_n \xrightarrow{p} \theta$.

Consistency is a very desirable property of an estimator: we often use estimators that are biased in finite samples, but we rarely use estimators that are not consistent.

We are now ready to return to the sample mean. The **Law of Large Numbers** tells us that, if X_i is i.i.d., then the sample mean, $\bar{X}_n$, converges in probability to the true mean.

Theorem (Law of Large Numbers). Suppose that X_i is iid, with $\mathbb{E}[X_i] = \mu$. Then $\bar{X}_n \xrightarrow{p} \mu$.

This is a remarkable result. We discussed in the first week how the sample mean is an unbiased estimator. Now we know that not only is the sample mean unbiased, it is also consistent! Note however that the law of large numbers adds the additional assumption of i.i.d. draws of X_i .

The law of large numbers is a powerful result, and will be very useful for showing that other estimators are consistent. However, it does not tell us the distribution of the estimator, and so it does not allow us to do inference. For that, we will need our next result.

The Central Limit Theorem

We saw already that as the sample size grows large, the distribution of the sample mean converges to a point. But along the way, what is the distribution of the sample mean? Answering this question will be essential for inference: to do hypothesis testing, we need to be able to construct test statistics and compare them to the critical values from a reference distribution.

To better understand this question, we will begin by standardizing the sample mean: we subtract $\mathbb{E}[\bar{X}_n]$ and divide by $\sqrt{\text{Var}(\bar{X}_n)}$. We can think of this as recentering and zooming in to the distribution of the sample mean, so that we can see the shape of the distribution even while it converges to a point. Since the sample mean is unbiased, its expected value is just the population mean, μ , which is 0.5 for our uniform distribution on $[0, 1]$. The variance of the sample mean is $\frac{\sigma^2}{n}$, and for our uniform distribution, $\sigma^2 = \frac{1}{12}$. We thus have $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma} = \sqrt{n} \cdot \frac{\bar{X}_n - 0.5}{\sqrt{1/12}}$.

Figure 2 plots the simulated distribution of $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$, for $n = 2, n = 5, n = 10$, and $n = 50$. Each figure plots the histogram for draws between -3 and 3 , as well as the standard normal distribution for reference. We can see that for $n = 2$ there is meaningful deviation from the normal distribution, but as the sample size increases, the distribution very quickly starts to look like a normal distribution. This happens much faster than the mean converges to a point: at $n = 10$ the normal distribution looks quite accurate, and it is almost exact by $n = 50$.

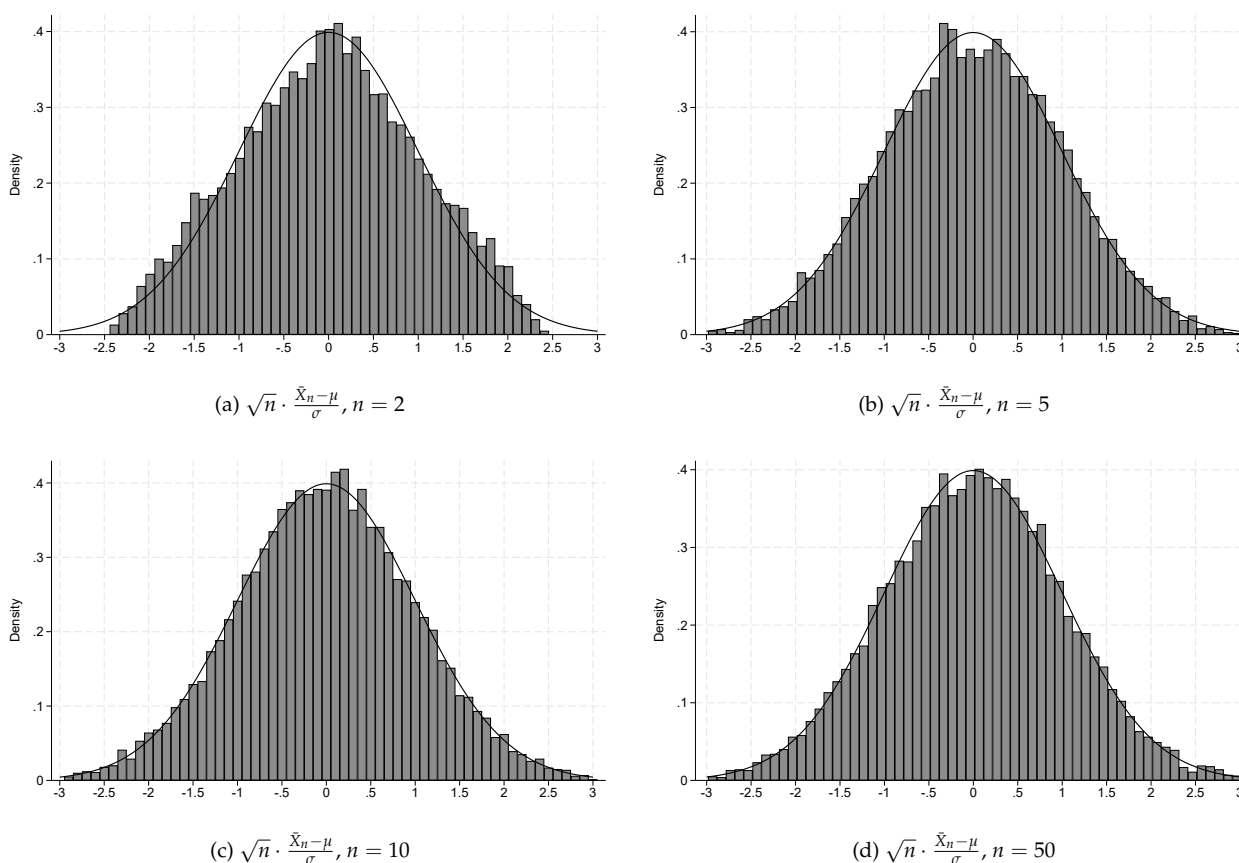


Figure 2: Distribution of Standardized Sample Mean, $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$, for Different Sample Sizes, n

Once again, it turns out that our simulation is conveying a more general property of the sample mean. To formalize this idea, we must first define **convergence in distribution**. Let $\{X_n\}_{n=1}^{\infty}$ denote a sequence of random variables, and let X denote a (continuous) random variable.

Formal Definition. We say that the sequence X_n converges in distribution to the continuous random variable X if the cdf of X_n converges to the cdf of X . That is, if X_n converges in distribution to X , then $\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$ for all x .

Intuitive Definition. We say that the sequence X_n converges in distribution to X if, as n grows large, the distribution of X_n gets arbitrarily close to the distribution of X .

We can write this as $X_n \xrightarrow{d} X$.

We can now state the **Central Limit Theorem**. This tells us that the sample mean is not only consistent, it is also asymptotically normal.

Theorem (Central Limit Theorem). Suppose that X_i is iid, with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$. Then $\sqrt{n} \cdot (\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$.

This is the most important result in statistics. The central limit theorem allows us to say something about the asymptotic distribution of the mean, even if we do not know the distribution of X_i . Moreover, since many estimators are themselves functions of sample means, the central limit theorem will allow us to derive the asymptotic distribution of other estimators, and thus let us do inference.

Once again, it is important to note that the CLT, as stated above, requires that sample draws be i.i.d. In cases where the sample is not i.i.d., it is often still possible to show asymptotic normality, but the variance will usually be different from the variance under i.i.d. draws.

The Continuous Mapping Theorem and Slutsky's Theorem

The LLN and CLT show us that the sample mean is consistent and asymptotically normal. This is great, but by itself is somewhat limited. What if we want to do statistics beyond just computing means?

It turns out that, in practice, most estimators can be expressed as functions of various sample means. We can use the CLT and LLN to show what each piece of the estimator converges to. And then we can use two additional tools, called the Continuous Mapping Theorem and Slutsky's Theorem, to show how the estimator behaves asymptotically.

If we know something about the convergence of X_n , what does that tell us about the convergence of $g(X_n)$, where g is some function? In traditional calculus, if $x_n \rightarrow x$ and g is continuous, then $g(x_n) \rightarrow g(x)$. It turns out that a similar notion applies for convergence in probability and in distribution.

Theorem (Continuous Mapping Theorem). Consider the function g , and the sequence $\{X_n\}_{n=1}^\infty$. Let X be a random variable, and let c be a constant.

- a) If $X_n \xrightarrow{d} X$ and g is continuous, then $g(X_n) \xrightarrow{d} g(X)$.
- b) If $X_n \xrightarrow{p} c$ and g is continuous at c , then $g(X_n) \xrightarrow{p} g(c)$.

The above theorem is known as the **Continuous Mapping Theorem**. A special case of this theorem is known as **Slutsky's Theorem**:

Theorem. Suppose that $X_n \xrightarrow{d} X$, and $Y_n \xrightarrow{p} y$, where y is a constant. Then:

- a) $X_n + Y_n \xrightarrow{d} X + y$.
 - b) $X_n \cdot Y_n \xrightarrow{d} y \cdot X$.
 - c) If $y \neq 0$, $X_n / Y_n \xrightarrow{d} X / y$.
- Suppose instead that $X_n \xrightarrow{p} x$, where x is a constant. Then:
- d) $X_n + Y_n \xrightarrow{p} x + y$.
 - e) $X_n \cdot Y_n \xrightarrow{p} y \cdot x$.
 - f) If $y \neq 0$, $X_n / Y_n \xrightarrow{p} x / y$.

The continuous mapping theorem and Slutsky's Theorem will be very useful to us when we wish to show that estimators are consistent and asymptotically normal. For example, suppose we have an estimator $\hat{\theta}_n = \frac{\hat{\beta}_n}{\hat{\gamma}_n}$, and we can show that $\hat{\beta}_n \xrightarrow{p} \beta$ and $\hat{\gamma}_n \xrightarrow{p} \gamma$. If $\theta = \frac{\beta}{\gamma}$, then this proves consistency. It turns out that our estimators often take this form, making Slutsky's Theorem extremely useful!

Exercise 1. Let μ and σ be constants. Suppose that $\sqrt{n} \cdot (\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$. Use Slutsky's Theorem to show that $\frac{\sqrt{n}}{\sigma} \cdot (\bar{X}_n - \mu) \xrightarrow{d} N(0, 1)$.

Exercise 2. Consider the setup of the preceding exercise. Suppose that $\hat{\sigma}$ is a consistent estimator of σ . Show that $\frac{\sqrt{n}}{\hat{\sigma}} \cdot (\bar{X}_n - \mu) \xrightarrow{d} N(0, 1)$.

Exercise 3. Use the above results and the definition of convergence in distribution to show that $\lim_{n \rightarrow \infty} P\left(\frac{\sqrt{n}}{\hat{\sigma}} \cdot (\bar{X}_n - \mu) > 2\right) = 1 - \Phi(2)$.

Exercise 4. Consider the above results in the context of the central limit theorem. How does this justify the use of t-tests?